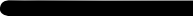
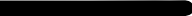


ICT AND SOCIETY

Part B: Digital Media and Multimedia



WHAT IS A DIGITAL IMAGE?

- A digital image is a representation of a two-dimensional image as a finite set of digital values, known as picture elements or pixels.
- Each pixel has a specific value that determines its color and brightness, and collectively, these pixels form the complete image.
- Digital images are the foundation of various technologies and applications in fields ranging from photography and medical imaging to remote sensing and computer vision.



TYPES OF DIGITAL IMAGE

Raster Images

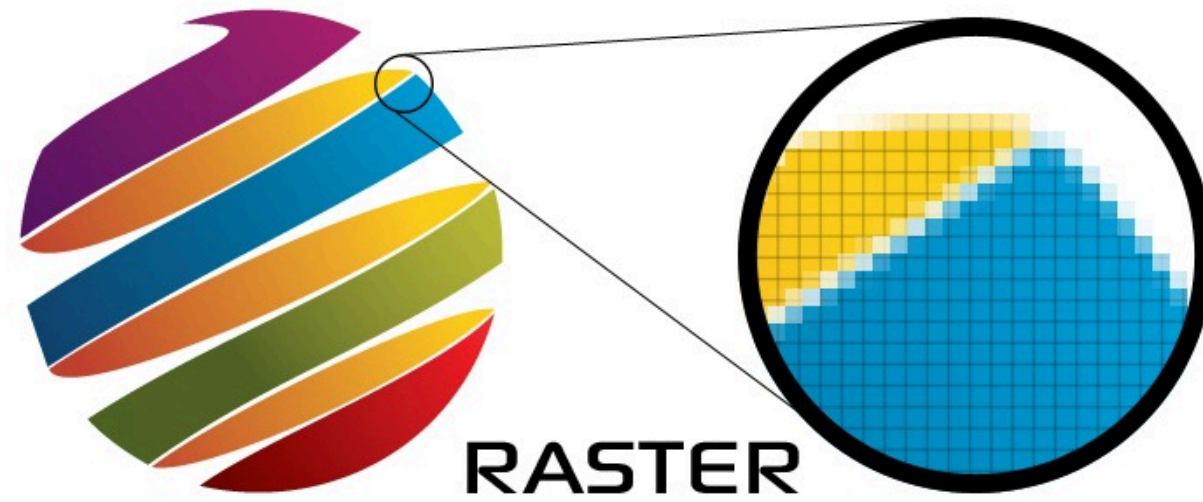
- Raster images, or bitmap images, are composed of a grid of individual pixels.
- Each pixel is assigned a specific color value.
- Raster images are resolution-dependent, meaning they can lose quality when scaled up.
- Common raster formats include JPEG, PNG, and GIF.

Vector Images

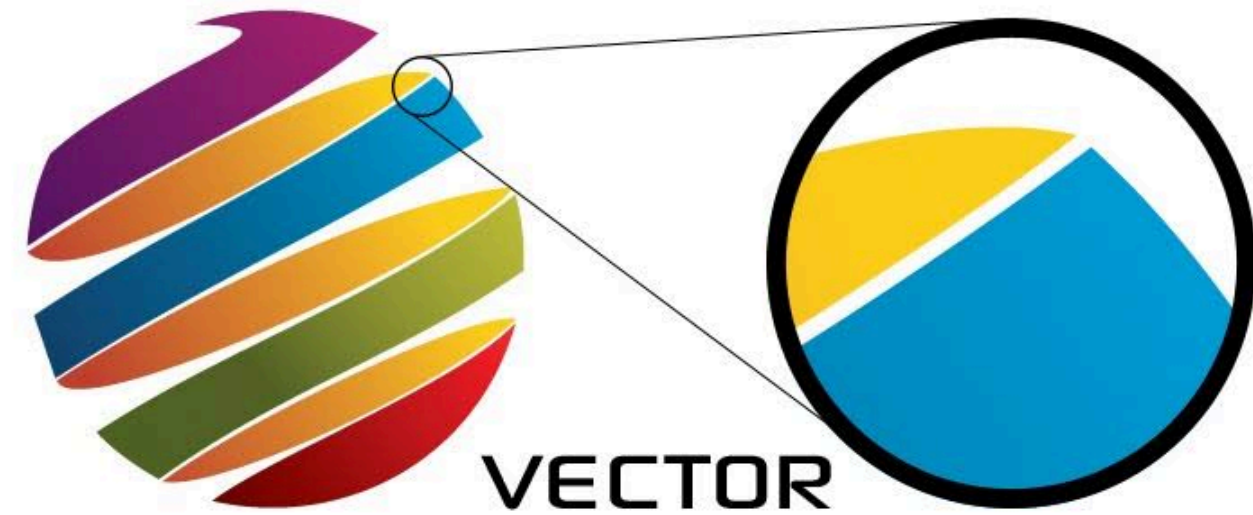
- Vector images use mathematical equations to represent shapes and lines, rather than individual pixels.
- They are resolution-independent, meaning they can be scaled to any size without losing quality.
- Common vector formats include SVG, EPS, and PDF. Vector images are often used for logos, illustrations, and typography.

EXAMPLE IMAGE

Raster Images



Vector Images



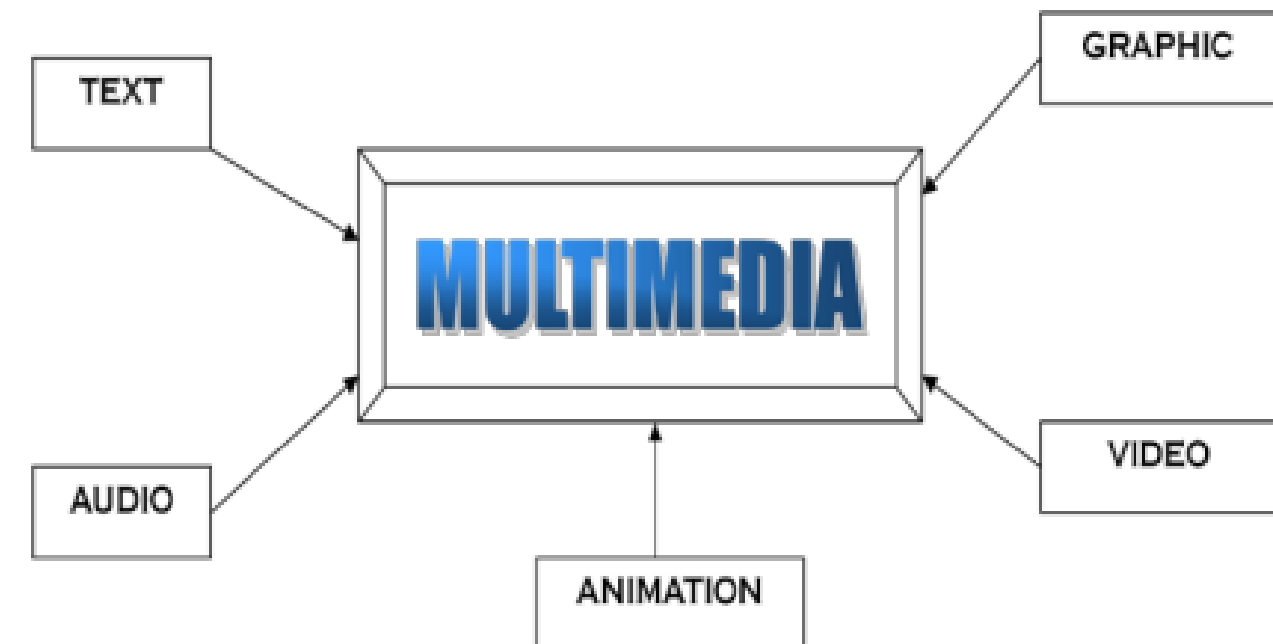
IS A GRAPHIC MULTIMEDIA?

Graphic

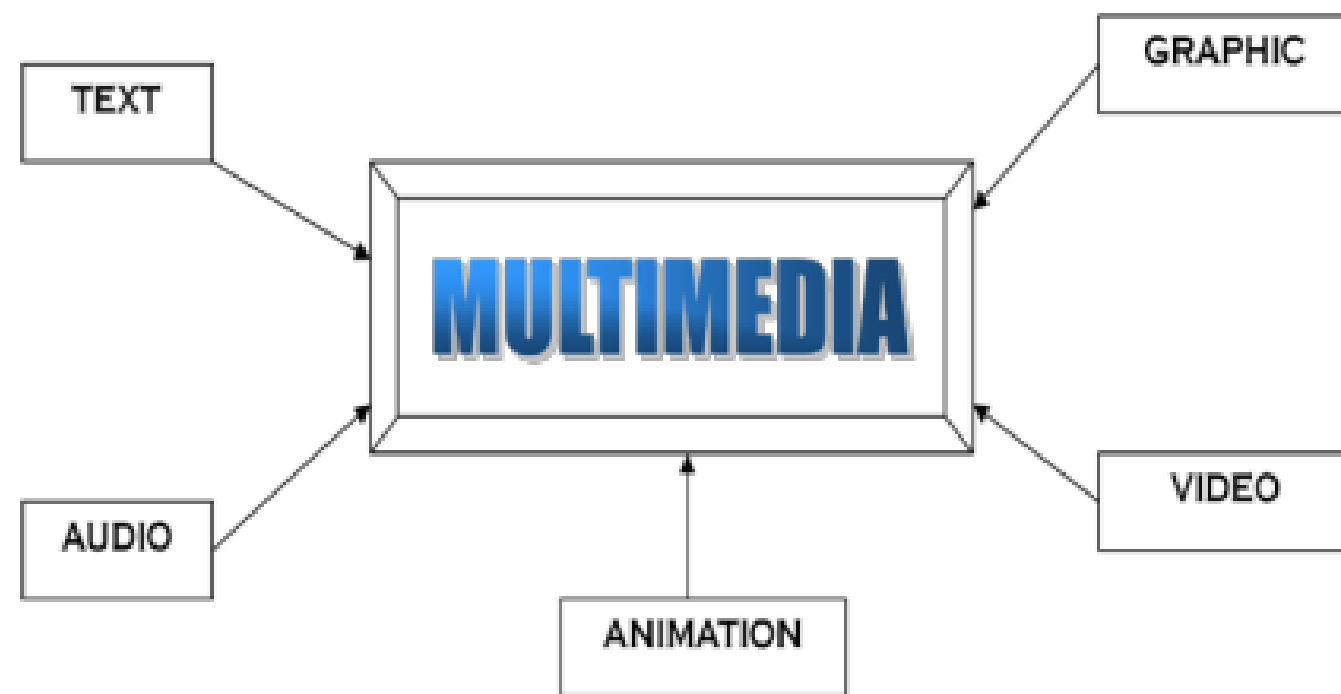
- Graphics refers to any visual representation of data and includes a variety of forms including drawings, photographs, line art, graphs, diagrams, numbers, and etc.
- Graphics may be images only or may also include text and they can be used for decorative or functional purposes, including educating, informing, illustrating, or entertaining.

Multimedia

- Is a combination of text, graphic, sound, animation, and video that is delivered interactively to the user by electronic or digitally manipulated means.



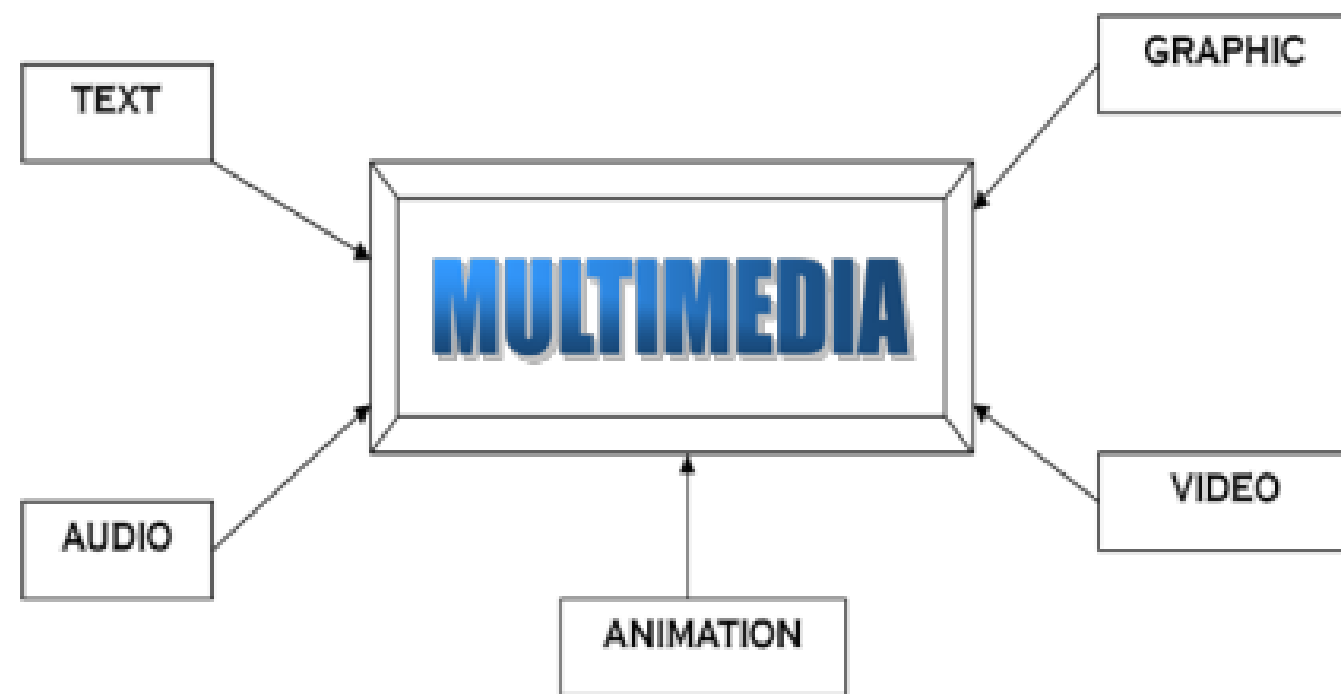
WHAT MAKES SOMETHING MULTIMEDIA?



Text

- A broad term for something that contains words to express something.
- Text is the most basic element of multimedia.
- A good choice of words could help convey the intended message to the users (keywords).
- Used in contents, menus, navigational buttons

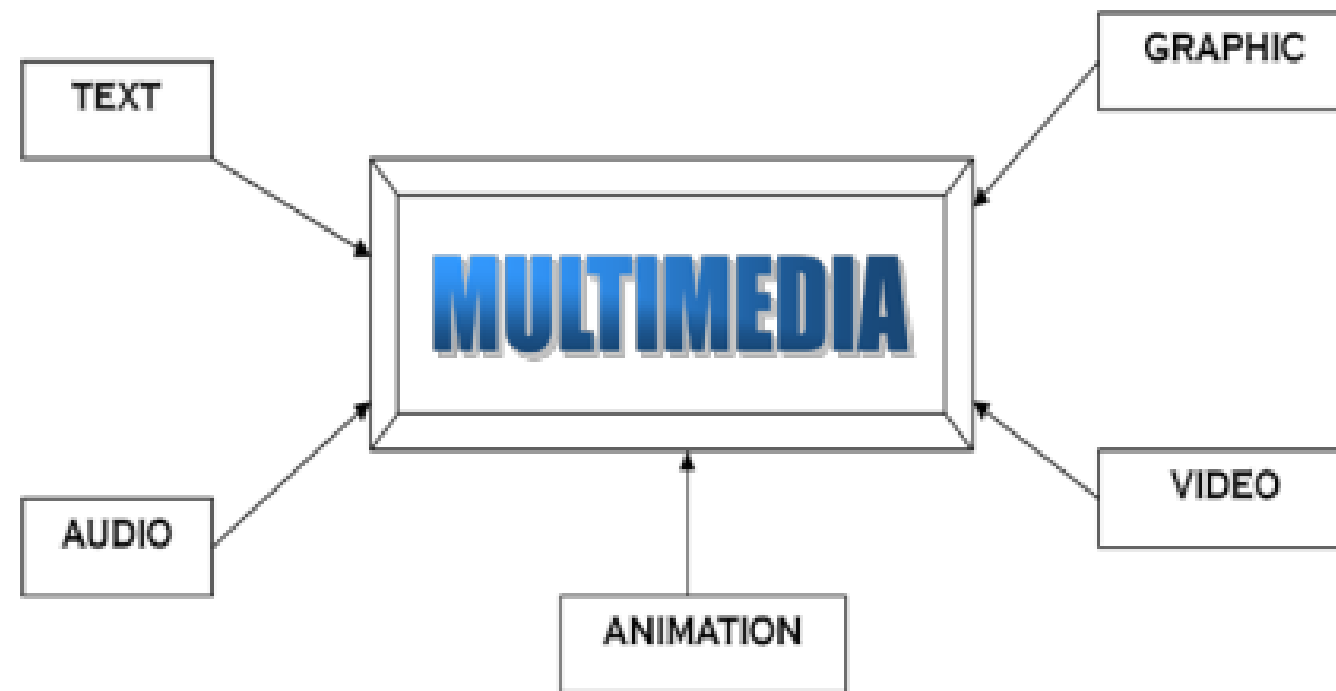
WHAT MAKES SOMETHING MULTIMEDIA?



Graphics

- Two-dimensional figure or illustration
- Could be produced manually (by drawing, painting, carving, etc.) or by computer graphics technology.
- Used in multimedia to show more clearly what a particular information is all about (diagrams, picture)

WHAT MAKES SOMETHING MULTIMEDIA?



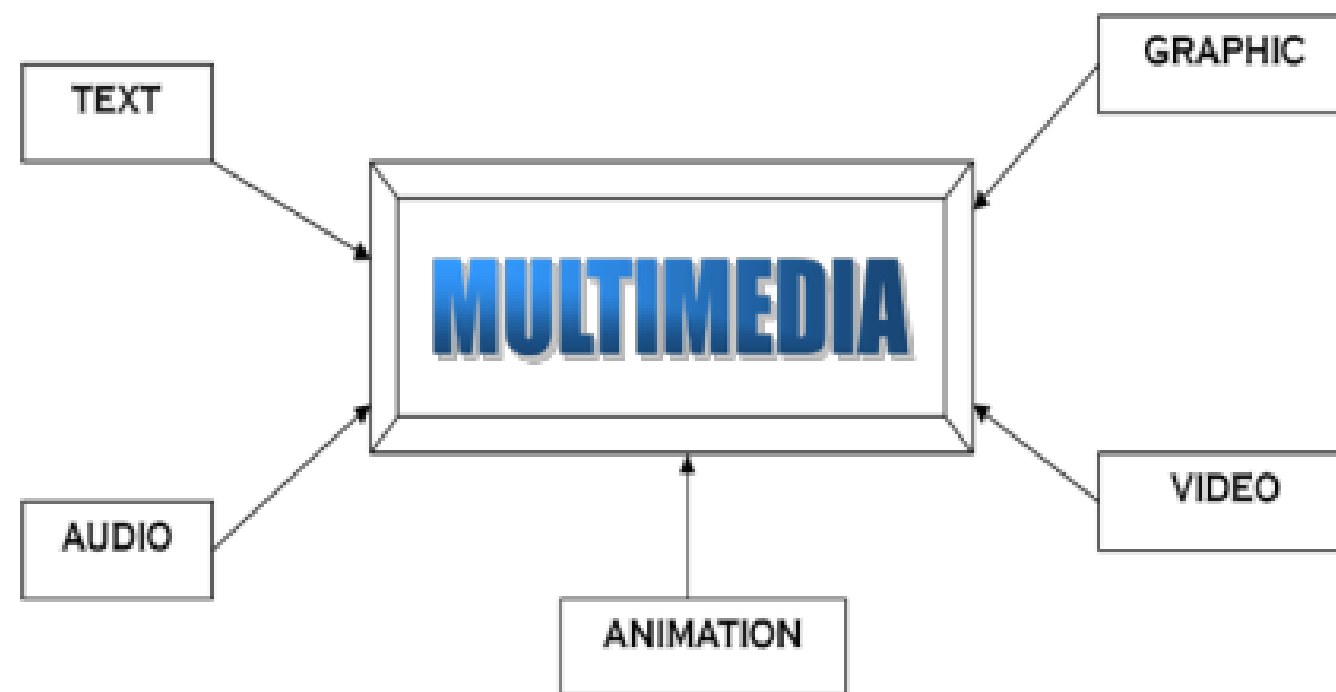
Audio

- Produced by vibration, as perceived by the sense of hearing.
- In multimedia, audio could come in the form of speech, sound effects and also music score.

Animation

- The illusion of motion created by the consecutive display of images of static elements.
- In multimedia, animation is used to further enhance / enriched the experience of the user to further understand the information conveyed to them.

WHAT MAKES SOMETHING MULTIMEDIA?



Video

- Video is the technology of capturing, recording, processing, transmitting, and reconstructing moving pictures.
- Video concentrates more on photo realistic image sequence and live recording.
- Video also takes a lot of storage space.
- It broadly divided into two types:
 - Linear
 - Non-linear

GRAPHICS VS. MULTIMEDIA

Graphics	Multimedia
Primarily visual	Combines multiple media
Image/illustration	Text + graphics + audio + video + animation, etc.
Usually static	May be dynamic/interactive
Example: logo	Example: interactive presentation

WHAT IS A MULTIMEDIA APPLICATION?

- Is a software program or system that combines two or more different types of media, such as text, graphics, audio, video, and animation, into an integrated digital experience, often with interactive controls for the user.
- Types of Multimedia Applications:
 - Graphics Applications
 - Audio Applications
 - Video Applications
 - Presentation and Authoring Software



GRAPHICS APPLICATIONS

Purpose

Create, edit, enhance, or manipulate digital images and graphics.

Examples:

- Adobe Photoshop → raster/photo editing
- Adobe Illustrator → vector graphics
- Canva → graphic design/content creation

AUDIO APPLICATIONS

Purpose

Record, edit, mix, and process digital sound.

Examples:

- Audacity
- Adobe Audition
- GarageBand

Possible tasks

- Recording narration, Cutting audio, Removing unwanted sections, Adjusting volume, Adding background music

VIDEO APPLICATIONS

Purpose

Create and edit moving visual content.

Examples:

- Adobe Premiere Pro
- DaVinci Resolve
- CapCut

Functions

- Cut/trim video, Combine clips, Add text, Add audio, Add transitions, Export video

PRESENTATION AND AUTHORIZING SOFTWARE

Purpose

Combine multiple media elements into a coherent presentation or interactive experience.

Examples:

- PowerPoint
- Google Slides
- Adobe Animate
- Multimedia authoring tools

WHICH SOFTWARE SHOULD I USE?

Task	Appropriate Application
Remove unwanted photo background	Image editor
Create scalable logo	Vector graphics software
Remove noise from narration	Audio editor
Cut a video	Video editor
Create a presentation	Presentation software
Create an interactive animation	Animation/authoring software

THE END